Muscle Proteomic Response to Blood Flow Restriction versus High-Load Resistance Training

Preliminary Analysis Report

Prepared for: **Dr. Cleiton Libardi**

Prepared by: **Sam Norton and Tahran Gotla**

August 07, 2026

This report summarizes data preprocessing, quality control, normalization, differential abundance, and pathway enrichment analyses completed to date for the BFR vs. HLRT muscle proteomics study. Results are preliminary and subject to revision as the analysis pipeline continues.

Contents

Project Overview

An examination of the proteomic alterations induced by blood flow restriction (BFR) vs. high-load resistance training (HLRT) in human vastus lateralis biopsy samples. N=33 participants' legs were randomly assigned to BFR or HLRT. Biopsies were donated PRE and POST training.

Proteomic analysis design at a glance:

Comparison	Contrast	Biological question
Baseline	BFR_PRE vs HLRT_PRE	Do the two arms differ before training begins?
Training_BFR	BFR_POST vs BFR_PRE	How does the muscle proteome change with BFR training?
Training_HLRT	HLRT_POST vs HLRT_PRE	How does the muscle proteome change with HLRT training?
Post_training	BFR_POST vs HLRT_POST	Do the two arms differ after training?
Interaction	(BFR_POST-BFR_PRE) vs (HLRT_POST-HLRT_PRE)	Does the training response itself differ by modality?

Table 2: Final sample counts per group after filtering and outlier removal: paired (both PRE and POST present), PRE-only, and POST-only subjects, with subject and sample totals.

dataset	group	n_paired	n_pre_only	n_post_only	n_subjects	n_samples
BFR	BFR	31	1	1	33	64
BFR	HLRT	32	1	0	33	65
BFR	Total	63	2	1	66	129

Pipeline Overview

The analysis pipeline proceeds through the following stages, each detailed in the sections below:

1. **Data preprocessing** — raw intensity/metadata loading, annotation cleanup, rerun consolidation
2. **Contaminant assessment** — tissue-specificity (HPA) and blood-protein flagging
3. **Filtering** — contaminant removal and per-group missingness filtering
4. **Outlier detection** — consensus-based sample QC
5. **Normalization** — cyclic loess normalization
6. **Differential abundance analysis** — limma, 5 contrasts
7. **Pathway / enrichment analysis** — GSEA + ORA across 7 databases, with redundancy filtering

1. Data Preprocessing

Raw protein-group intensities and sample metadata were loaded and restructured prior to any filtering:

- Protein annotation columns (UniProt ID, protein name, gene symbol, description) were parsed from the search-engine output, and multi-protein group matches were collapsed to their first (primary) match.
- cRAP (common Repository of Adventitious Proteins) contaminant tags were flagged directly from the raw protein group string, before any trimming could drop that information.
- Sample columns were cleaned to a consistent `S###` naming convention.
- **Rerun consolidation:** samples with a trailing "r"/"R" suffix (indicating a repeat MS injection) were identified and merged onto their base sample name, with a `rerun_level` flag retained in the metadata for traceability.
- Metadata and intensity columns were validated to be in 1:1 correspondence before proceeding (a hard `stopifnot` check — the pipeline halts rather than silently misaligning samples).

2. Contaminant Assessment

Two independent contamination checks were applied to every protein, in addition to a per-sample contamination index:

A. Tissue specificity (Human Protein Atlas). Each protein's UniProt ID was checked against the HPA skeletal muscle gene list; proteins absent from this reference were flagged as not muscle-specific.

B. Blood protein blacklist. A curated list of classic blood/plasma contaminant gene families was applied — albumin & lipid transport proteins, coagulation factors, complement system components, hemoglobins, acute-phase proteins, and platelet/neutrophil markers — supplemented dynamically with any immunoglobulin (IGH/IGK/IGL) or keratin (KRT) genes detected in this specific dataset.

C. Sample-level contamination index. For every sample, the summed intensity of blood markers (ALB, HBB, HBA1) vs. muscle markers (MB, CKM) was expressed as a percentage of total intensity, giving a blood:muscle (B:M) ratio per sample — a diagnostic for samples with unusually high blood contamination (e.g. from incomplete blood clearance during biopsy).

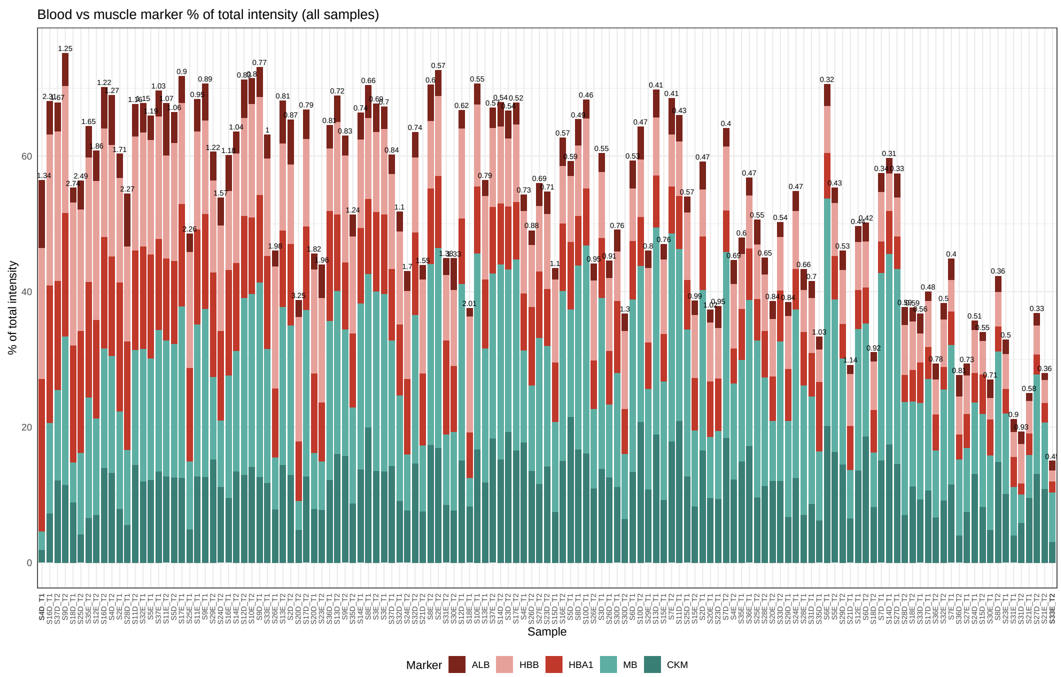

Figure 1: Blood vs. muscle marker proportion of total intensity, per sample.

3. Filtering

Proteins and samples were filtered in the following order:

1. **Zero → missing.** Raw zero-intensity values (an artifact of the search engine, not a true measurement of zero abundance) were converted to `NA`.
2. **HPA muscle filter.** Proteins not annotated as muscle-expressed in the Human Protein Atlas were removed.
3. **Blood blacklist filter.** Proteins matching the blood contaminant gene list (Section 2B) were removed.
4. **Missingness filter.** Proteins were required to be observed in at least 50% of samples *within* at least one group (BFR or HLRT), rather than 50% of the full cohort — this retains proteins that are consistently detected in one arm even if largely absent in the other, which a whole-cohort threshold would have discarded.

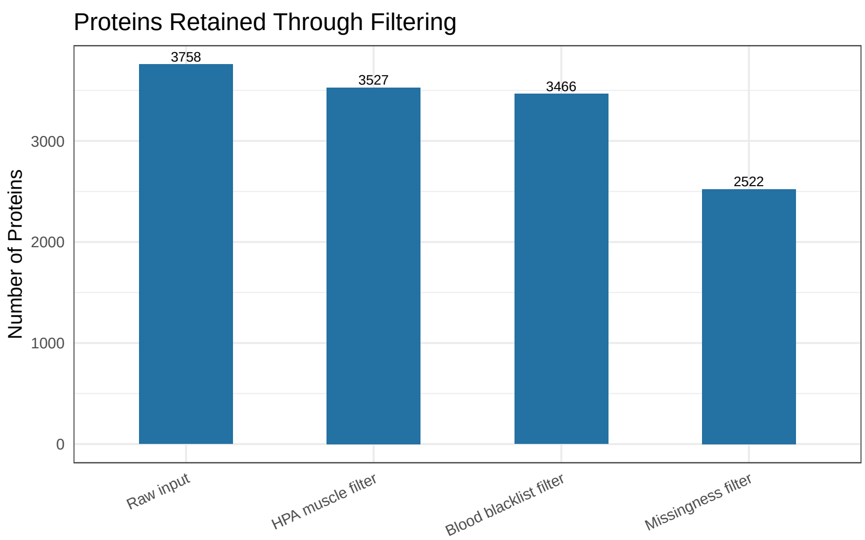

Figure 2: Number of proteins retained at each filtering step.

Table 3: Filtering effects: proteins before/after each step.

step	n_before	n_after	n_removed	pct_retained
Raw input	NA	3758	NA	100.0
HPA muscle filter	3758	3527	231	93.9
Blood blacklist filter	3527	3466	61	92.2
Missingness filter	3466	2522	944	67.1

4. Outlier Detection

Samples were screened for exclusion using a **consensus of three independent methods**, each flagging samples that are statistical outliers relative to the rest of the cohort:

- **Missingness outliers:** samples with % missing proteins beyond 1.5×IQR above the 75th percentile.
- **PCA outliers:** samples with a Mahalanobis distance (computed on the first 3 principal components of log2 intensities) exceeding the 99th-percentile χ^2 threshold.
- **Intensity outliers (MAD):** samples whose median log2 intensity deviates from the cohort median by more than 3× the median absolute deviation (MAD).

A sample was excluded only if flagged by **at least 2 of the 3** methods, to avoid removing samples based on a single noisy metric.

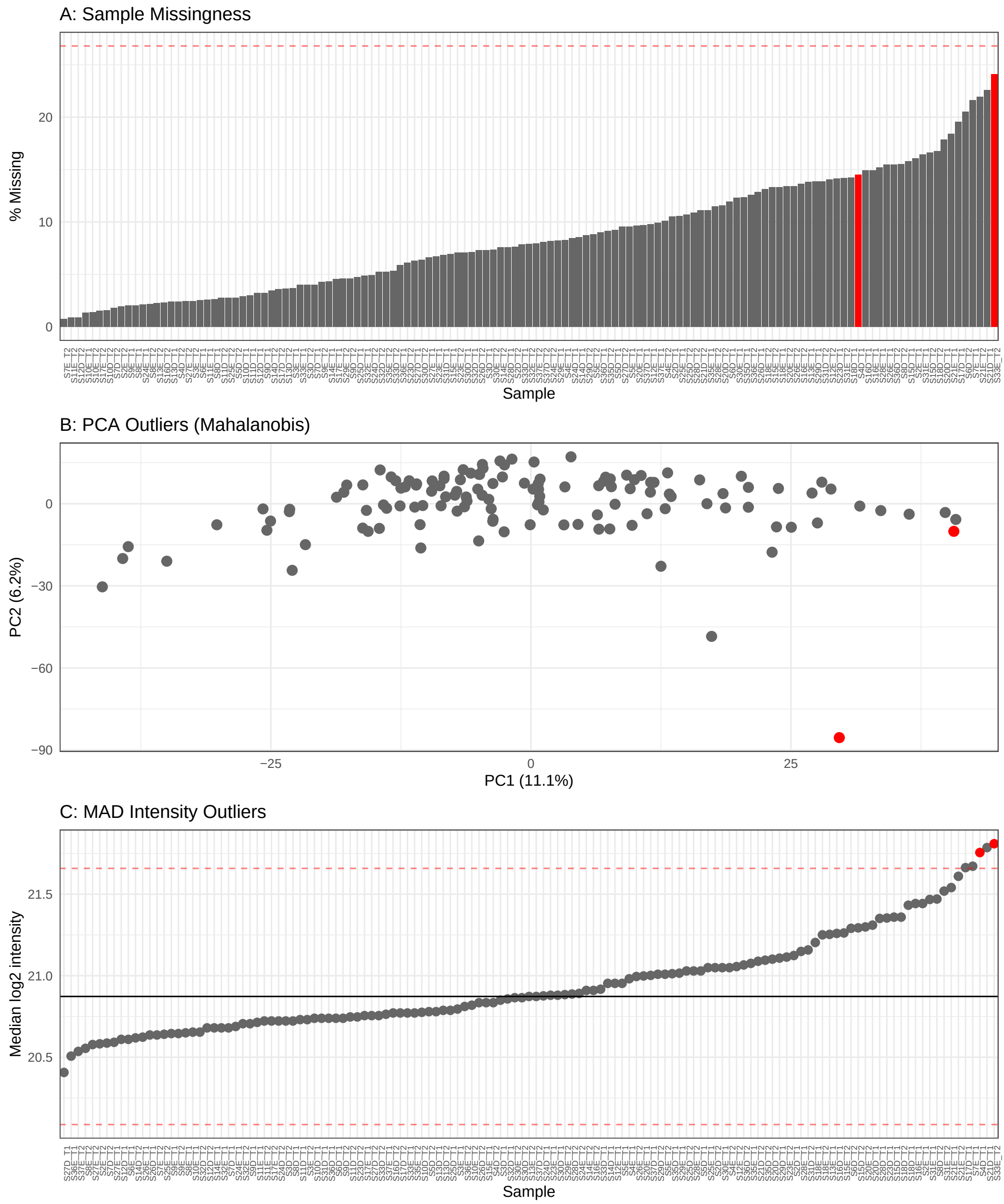

Figure 3: Outlier diagnostics: missingness, PCA (Mahalanobis), and MAD intensity.

5. Normalization

Following filtering and outlier removal, protein intensities were normalized using **cyclic loess normalization**, which iteratively aligns the intensity distribution of each sample against the others via local regression (loess) on pairwise MA-plots. This corrects for sample-to-sample technical variation (e.g. differences in total protein loaded, digestion/labeling efficiency) without assuming the majority of proteins are unchanged between the two study arms, which a simpler median/quantile normalization would.

Quality control before vs. after normalization:

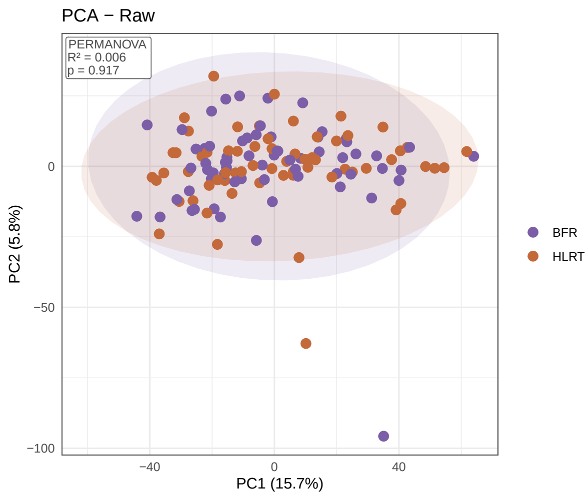

Figure 4: PCA of raw (pre-normalization) intensities, colored by group.

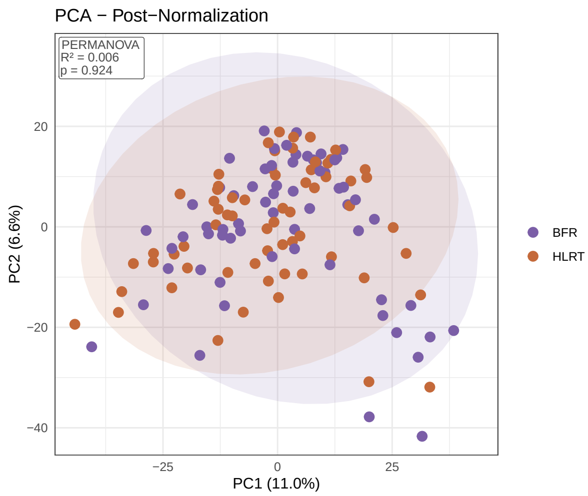

Figure 5: PCA of normalized intensities, colored by group.

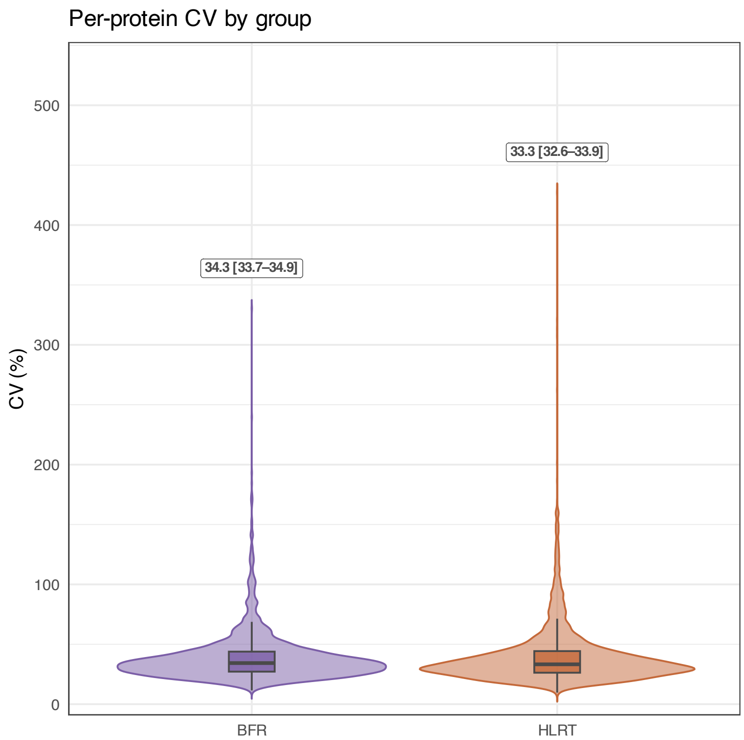

Figure 6: Per-protein coefficient of variation (CV) by group, pre-normalization.

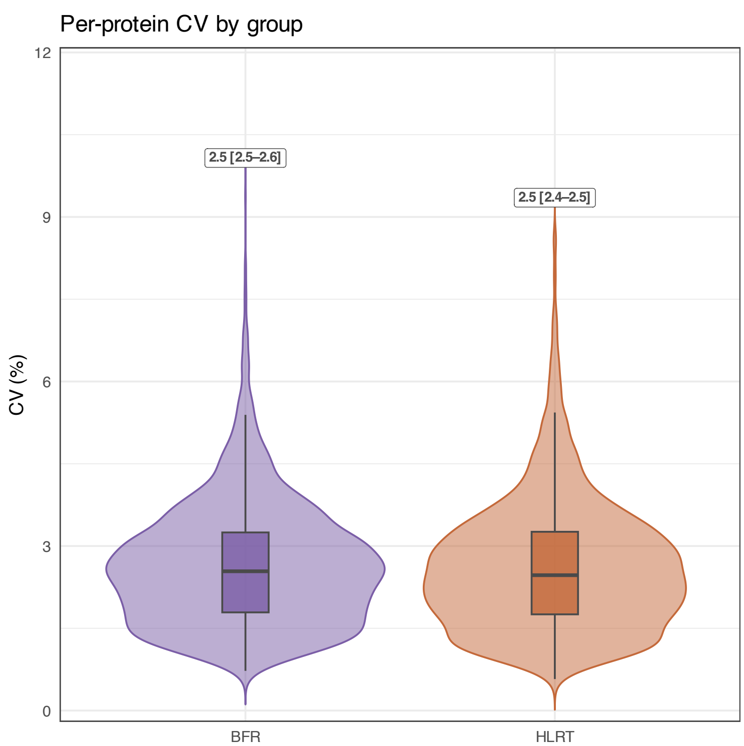

Figure 7: Per-protein coefficient of variation (CV) by group, post-normalization.

6. Differential Abundance Analysis

Differential abundance was assessed with **limma**, fit across 5 contrasts representing the study's 2 × 2 (arm × training) design plus the arm×time interaction (see the contrast table in Section "Project Overview"). For each contrast, three complementary significance metrics were retained: the raw p-value, the Benjamini-Hochberg FDR-adjusted p-value, and a composite **Pi-value** (p^|log2FC|; Xiao et al., 2014) used as the primary threshold (Pi-value < 0.05) for calling a protein differentially abundant.

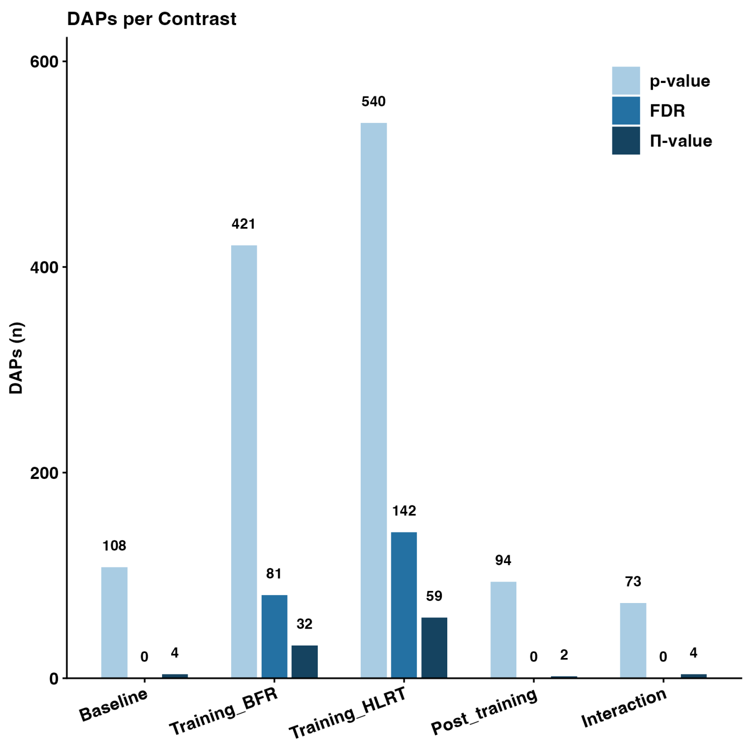

Figure 8: Number of differentially abundant proteins (DAPs) per contrast, by significance method.

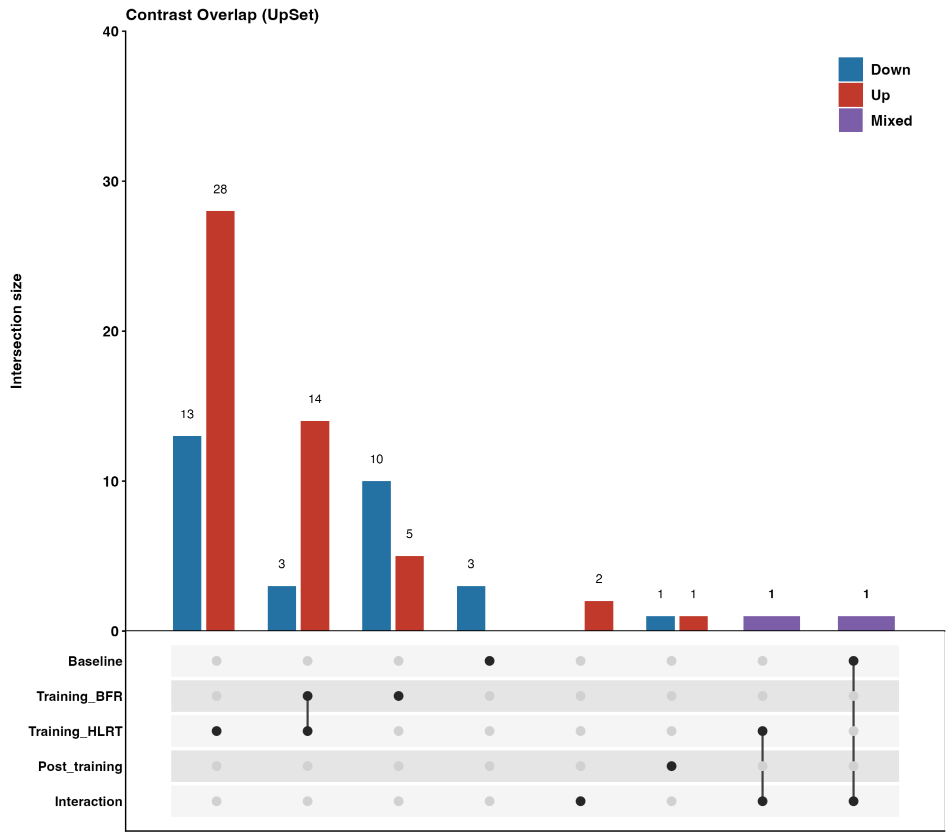

Figure 9: Overlap of differentially abundant proteins across contrasts (UpSet plot), colored by direction of change.

Volcano Plots by Contrast

Per-protein volcano plots for each of the 5 contrasts, showing effect size (log2 fold change) against significance (-log10 p-value). Proteins are colored by direction of change, with the top proteins by fold-change magnitude labeled in each panel and UP/DOWN counts annotated in the corners.

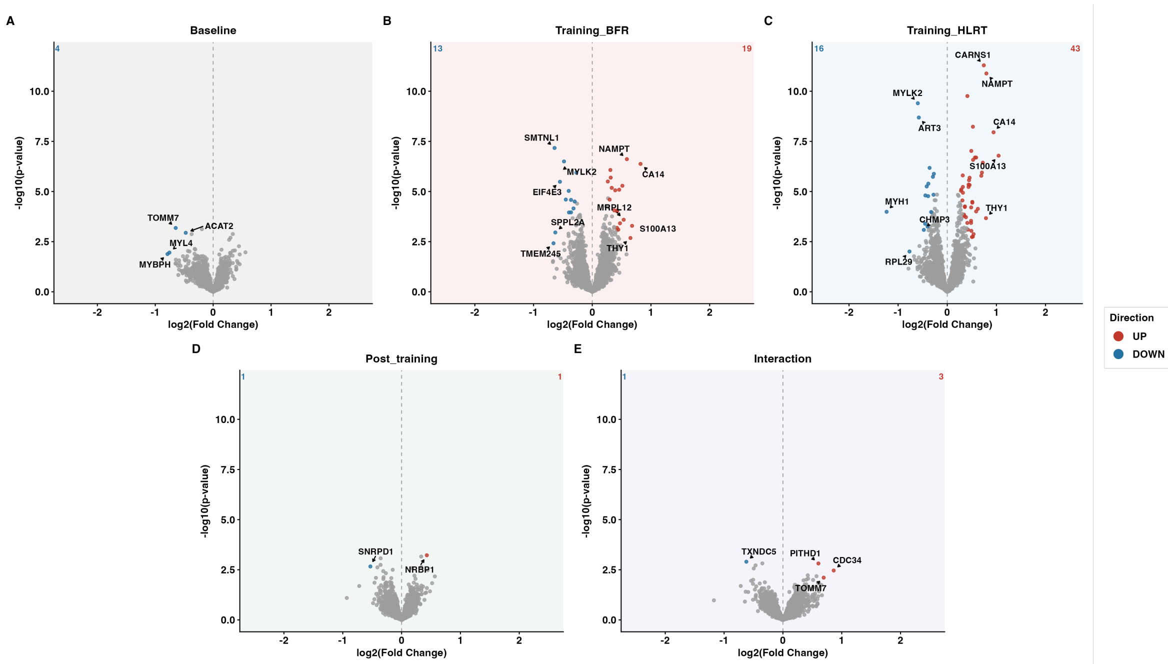

Figure 10: Volcano plots for all 5 contrasts.

Protein-Level Intensity Plots

Proteins concordantly significant in both contrasts. The following show proteins that were significantly up- or down-regulated with the *same direction* in both Training_BFR and Training_HLRT (ranked so both contrasts' p-values must be small, not just one), with all four group × timepoint combinations shown per protein (BFR PRE/POST, HLRT PRE/POST) to visualize the shared response pattern across both training modalities directly.

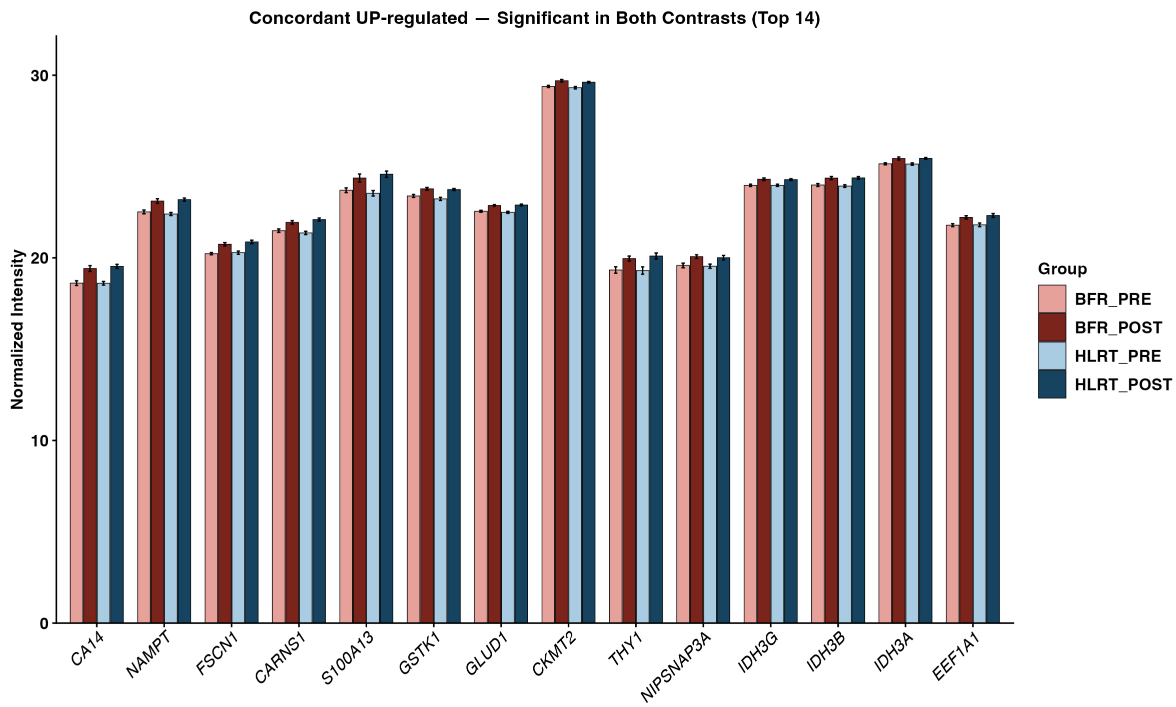

Figure 11: Top 20 up-regulated proteins, significant in both Training_BFR and Training_HLRT.

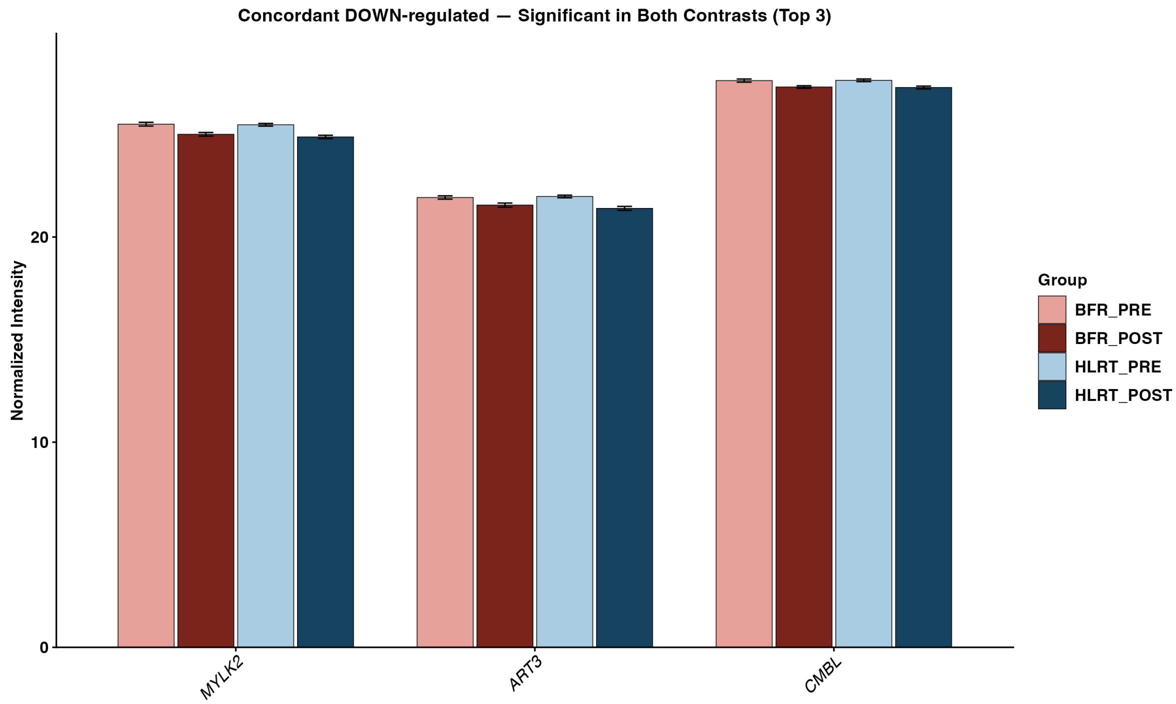

Figure 12: Top 20 down-regulated proteins, significant in both Training_BFR and Training_HLRT.

7. Pathway / Enrichment Analysis

For each of the 5 contrasts, two complementary enrichment approaches were run:

Gene Set Enrichment Analysis (GSEA) — rank-based, using the limma t-statistic as the ranking metric, tested against 7 pathway/gene-set databases: GO Biological Process, GO Cellular Component, GO Molecular Function (via `clusterProfiler::gseGO`), a GO Slim subset, KEGG, Reactome, and MSigDB Hallmark gene sets.

Over-Representation Analysis (ORA) — tested the sets of significantly up- and down-regulated proteins (Pi-value < 0.05) separately against GO BP/CC/MF, using all detected proteins as the statistical background. NOTE: no ORA results are included in this report.

Redundancy filtering. Because gene set databases contain many near-duplicate terms (e.g. highly overlapping GO terms), a pooled greedy redundancy filter was applied within each contrast $\times$ database: significant pathways were ranked by adjusted p-value, and a pathway was dropped if its gene-set membership overlapped an already-retained, more-significant pathway above a threshold (Jaccard index ≥ 0.5 **or** overlap coefficient ≥ 0.5). This keeps the most significant representative of each cluster of redundant terms rather than reporting near-duplicates as independent hits.

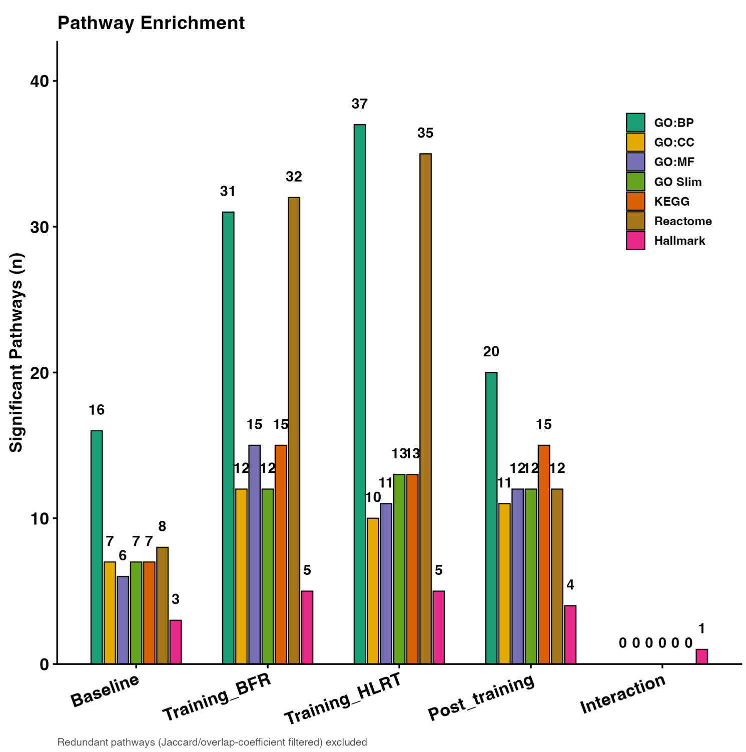

Figure 13: Number of significant (non-redundant) enriched pathways per contrast, by database.

GSEA Bubble Plot

Significant Hallmark and GO Biological Process pathways for the Training_BFR and Training_HLRT contrasts, restricted to redundancy-filter representative pathways (Section 7's redundancy filtering, applied above before ranking). Point size reflects statistical confidence (-$\log_{10}$

FDR), fill color reflects normalized enrichment score (NES, red = up in that contrast, blue = down), and outline indicates whether that specific point clears the FDR < 0.05 threshold.

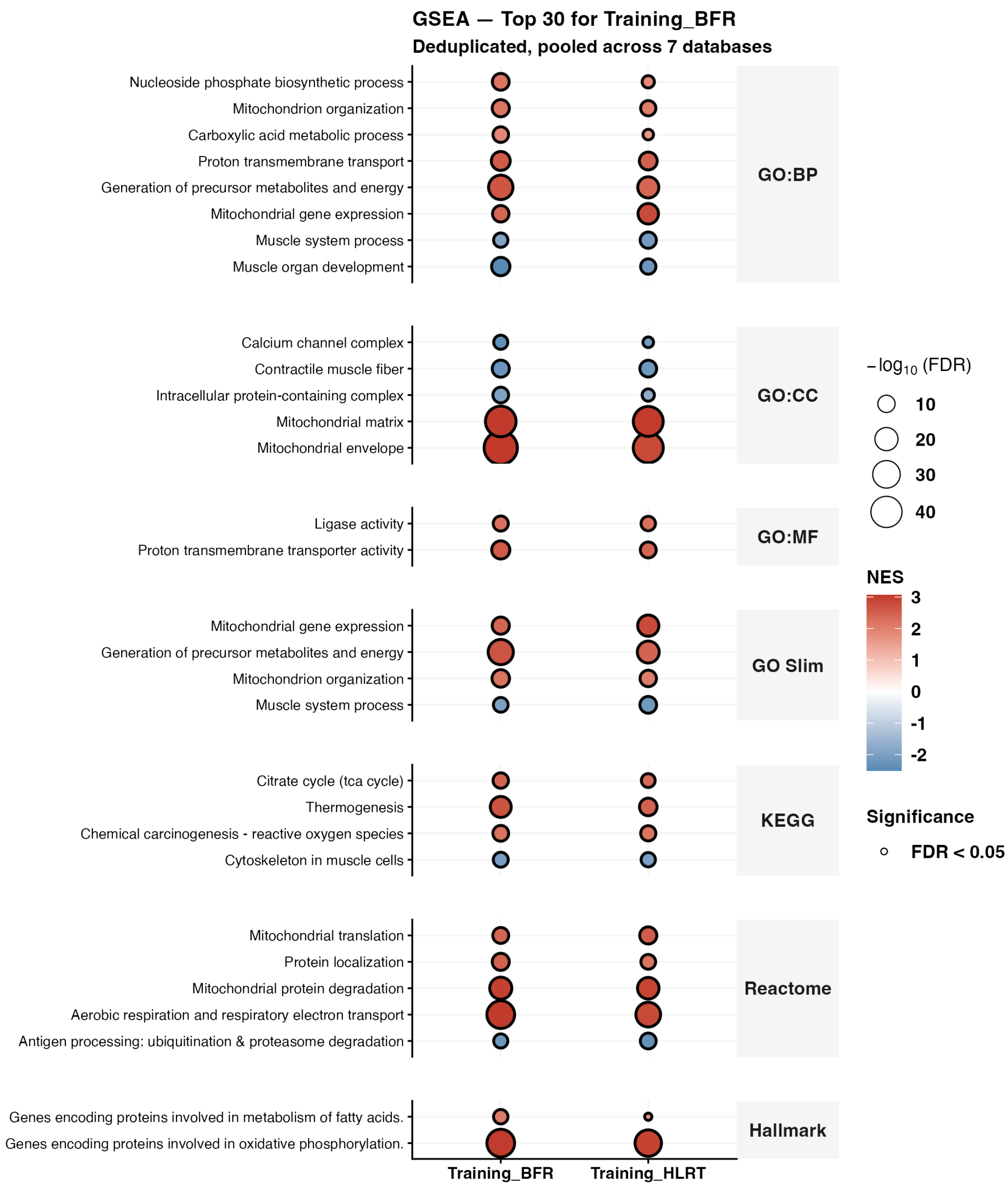

Figure 14: GSEA bubble plot: Top 30 pathways significant in the Training_BFR contrast across 7 databases.

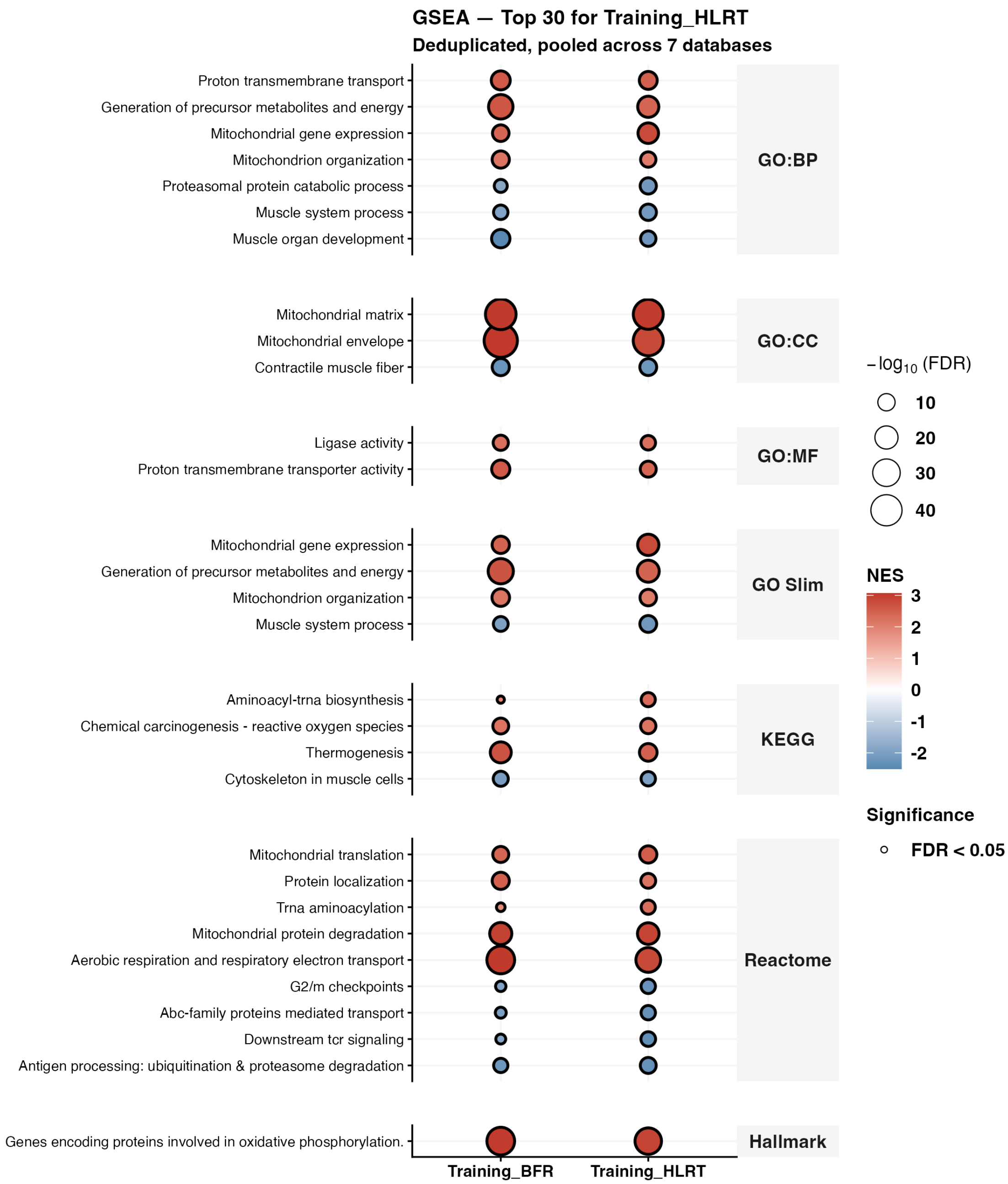

Figure 15: GSEA bubble plot: Top 30 pathways significant in the Training_HLRT contrast across 7 databases.

Differential Pathway Analysis: BFR vs. HLRT Training Response

To assess whether the two training modalities engage the same or different biological pathways, every redundancy-filtered pathway was classified by its significance and direction (NES sign) in Training_BFR vs. Training_HLRT: **BFR only** or **HLRT only** (significant in one contrast but not the other), **Discordant** (significant in both, but opposite direction), or **Concordant** (significant in both, same direction). Pathways in the first three categories represent genuine differences between the two training modalities; concordant pathways are explicitly excluded from that "different" set, since those are pathways that respond the same way regardless of modality. NOTE: no significant discordant pathways were identified.

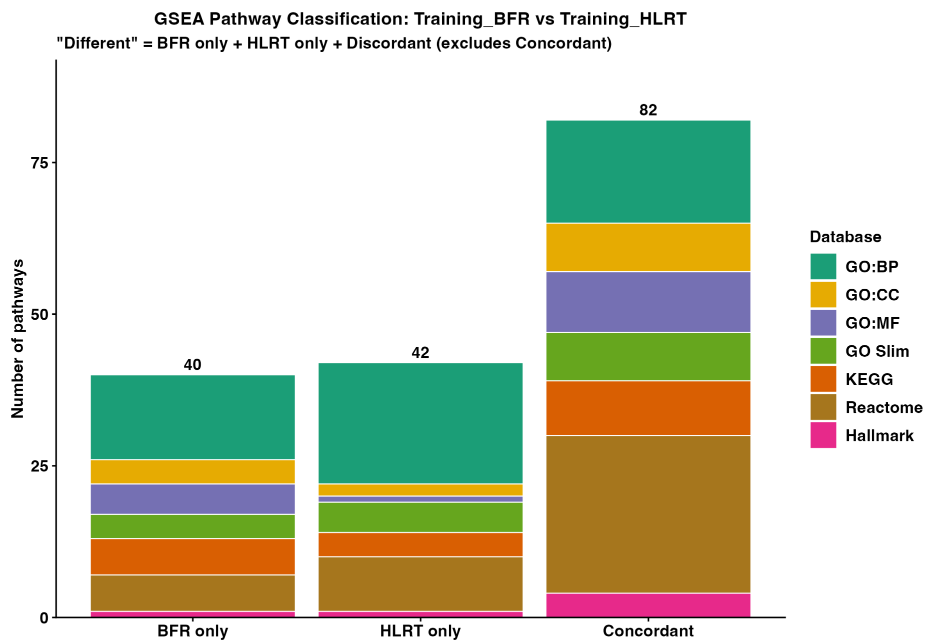

Figure 16: Pathway classification counts by status and database, Training_BFR vs Training_HLRT.